Christopher Chen

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Education

Brown University Providence, RI

Bachelor of Science in Computer Science and Applied Mathematics *September 2022 – May 2026*

- **Relevant Coursework**: Design and Analysis of Algorithms, Operating Systems, Deep Learning, Computer Networks, Database Management Systems, Compilers, Numerical Optimization, Operations Research
- **Teaching Assistant**: Discrete Structures and Probability (CSCI 0220)

Experience

GitHub June 2025 – August 2025

Software Engineering Intern *Remote*

- Designed and deployed 15+ RESTful API endpoints in Ruby on Rails, architected to serve 1M+ GitHub Projects upon full release with OAuth and access control systems
- Implemented agentic AI features for GitHub Projects via MCP server integration, enabling autonomous workflows targeting GitHub Copilot adoption
- Engineered a bidirectional cursor-based pagination system and optimized N+1 query performance through strategic association preloading, improving API response times by 60%+

Hightainment (Startup) January 2025 – April 2025

Software Engineering Intern *Remote*

- Built Python APIs and reusable React components in an Agile startup environment, improving server scalability and ensuring consistent user experience
- Automated workflows with AWS Lambda pipelines, integrating public APIs with custom scraping to replace daily manual data processing for entertainment media
- Optimized MySQL schemas with better indexing, reducing query overhead cutting API response times by 20%

Vane (Startup) June 2024 – October 2024

Software Engineering Intern *Remote*

- Developed core iOS app components with Flutter/Dart, collaborating with designers in Figma to implement friend addition, event search, and user profile features
- Built a full-stack database editor tool with React and Node.js, streamlining internal data management workflows
- Assisted in migrating backend from MongoDB to Supabase, designing schemas and APIs to support faster queries

Projects

Fliptastic | *Python, React, OpenCV, FastAPI* April 2025 – May 2025

- Engineered a multimodal computer vision pipeline using MediaPipe Face Mesh and OpenCV to detect blinks, gaze shifts, and head tilts as means for hands-free PDF navigation with up to 100% accuracy in trials
- Built a full-stack web app to integrate real-time webcam gesture processing with cross-device PDF page flipping

IP/TCP | *Rust* October 2024 – December 2024

- Designed a concurrent TCP/IP network stack in Rust implementing reliable data transfer with retransmission mechanisms, achieving successful communication with 2% packet loss tolerance

DBMS | *Go* September 2024 – December 2024

- Engineered an ACID-compliant, thread-safe database management system in Go with B+ tree and extendible hash indexing for data retrieval
- Implemented logical query optimization, bloom filters, and a write-ahead log (WAL) for crash recovery

Weenix OS | *C* January 2024 – May 2024

- Developed a Unix-based kernel in C with preemptive scheduling supporting 1,000+ concurrent processes
- Built multi-threaded file systems with buffer caching, enabling high-throughput file operations via POSIX calls

Technical Skills

Languages: Python, C/C++, Java, Javascript, Ruby, Go, Rust, Dart
Frameworks & Libraries: Node, MongoDB, Express, React, Pandas, TensorFlow, PyTorch, NumPy, OAuth
Tools: Git, Docker, AWS, SQL, NoSQL, REST API, TailwindCSS, Figma
Concepts: Backend Development, Frontend Development, Fullstack Development, Machine Learning, CI/CD, Agile